

EXPERIMENT NO. 6

Deluge Programming in Zoho Creator

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AIM

To write and execute Deluge scripts in Zoho Creator for performing calculations, conditional operations, loops and database-related actions.

INTRODUCTION

Deluge (Data Enriched Language for the Universal Grid Environment) is the scripting language used in Zoho applications. It is used to automate tasks, manipulate form data, perform calculations, access records and execute actions such as sending notifications. The following programs cover basic Deluge statements as well as common Zoho Creator record operations.

PROGRAMS INCLUDED

No.	Program	Concept
1	DISPLAY A MESSAGE	info statement
2	ADD TWO NUMBERS	arithmetic
3	TOTAL AND AVERAGE	calculation
4	IF-ELSE CONDITION	if-else
5	FOR EACH LOOP	loop
6	INSERT A STUDENT RECORD	insert record
7	FETCH A RECORD	fetch record
8	UPDATE A RECORD	update record
9	CALCULATE POTENTIAL REVENUE	calculation
10	SEND AN EMAIL	sendmail

NOTE

Programs 6–8 assume that a form named **Students** exists with fields Student_Name, Register_Number, Department and Year. The email program uses the Zoho Creator mail action syntax; replace the sample recipient address with the required address when testing.

PROGRAM 1 — DISPLAY A MESSAGE

Purpose: Display a message using the info statement.

```
info "Welcome to the College Event Registration System";
```

Deluge Code

Output

```
Welcome to the College Event Registration System
```

PROGRAM 2 — ADD TWO NUMBERS

Purpose: Perform addition of two numeric values.

```
a = 25;
b = 15;
sum = a + b;
info "First Number : " + a;
info "Second Number: " + b;
info "Sum          : " + sum;
```

Deluge Code

Output

```
First Number : 25
Second Number: 15
Sum          : 40
```

PROGRAM 3 — TOTAL AND AVERAGE

Purpose: Find the total and average of three marks.

Deluge Code

```
m1 = 80;
m2 = 75;
m3 = 90;
total = m1 + m2 + m3;
average = total / 3;
info "Total   : " + total;
info "Average : " + average;
```

Output

```
Total   : 245
Average : 81.6666667
```

PROGRAM 4 — IF-ELSE CONDITION

Purpose: Check whether a mark satisfies the pass condition.

Deluge Code

```
marks = 72;
if(marks >= 50)
{
    info "Result : PASS";
}
else
{
    info "Result : FAIL";
}
```

Output

```
Result : PASS
```

PROGRAM 5 — FOR EACH LOOP

Purpose: Repeat a statement for a sequence of values.

```
for each n in 1..5
{
    info "Number : " + n;
}
```

Deluge Code

Output

```
Number : 1
Number : 2
Number : 3
Number : 4
Number : 5
```

PROGRAM 6 — INSERT A STUDENT RECORD

Purpose: Insert a new record into the Students form.

```
insert into Students
[
    Student_Name = "Aadhavan RP"
    Register_Number = "240701819"
    Department = "CSE"
    Year = "3rd Year"
];
```

Deluge Code

Output

```
Record inserted successfully
Student Name : Aadhavan RP
Register No   : 240701819
Department    : CSE
```

PROGRAM 7 — FETCH A RECORD

Purpose: Fetch a student record using a criteria condition.

Deluge Code

```
student = Students[Register_Number == "240701819"];
if(student.count() > 0)
{
    info "Student : " + student.Student_Name;
    info "Department : " + student.Department;
}
else
{
    info "Student record not found";
}
```

Output

```
Student : Aadhavan RP
Department : CSE
```

PROGRAM 8 — UPDATE A RECORD

Purpose: Update a field in an existing student record.

Deluge Code

```
student = Students[Register_Number == "240701819"];
if(student.count() > 0)
{
    student.Department = "CSE";
    info "Student record updated successfully";
    info "Department : " + student.Department;
}
```

Output

```
Student record updated successfully
Department : CSE
```

PROGRAM 9 — CALCULATE POTENTIAL REVENUE

Purpose: Calculate potential revenue from quantity and unit price.

Deluge Code

```
quantity = 5;
price = 1200;
potential_revenue = quantity * price;
info "Quantity      : " + quantity;
info "Price per Unit : " + price;
info "Potential Revenue : " + potential_revenue;
```

Output

```
Quantity      : 5
Price per Unit : 1200
Potential Revenue : 6000
```

PROGRAM 10 — SEND AN EMAIL

Purpose: Execute an email notification action.

Deluge Code

```
sendmail
[
    from : zoho.adminuserid
    to : "student@example.com"
    subject : "Event Registration Confirmation"
    message : "Your registration has been received successfully."
];
```

Output

```
Email action executed successfully
Subject : Event Registration Confirmation
Message : Your registration has been received successfully
```

RESULT

Thus, ten Deluge programs were written for Zoho Creator covering message display, arithmetic, conditional statements, loops, record insertion, record retrieval, record updating, calculation and email automation. The expected outputs were verified for each program.